AI Security

Introduction to Agentic AI

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POSTECH

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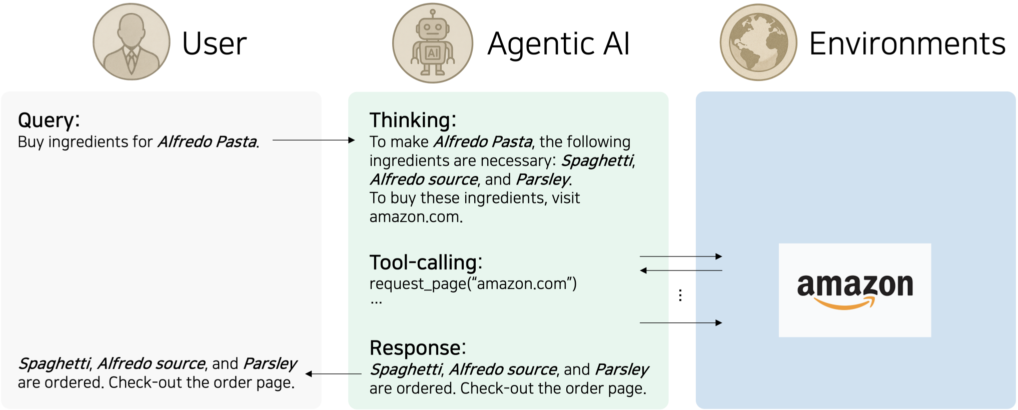
Outline

Era of Agentic AI



- What's under the hood? We will consider the following questions:
  - What's the core components and how do they work?
  - How to tame LLMs for Agentic AI?

Outline

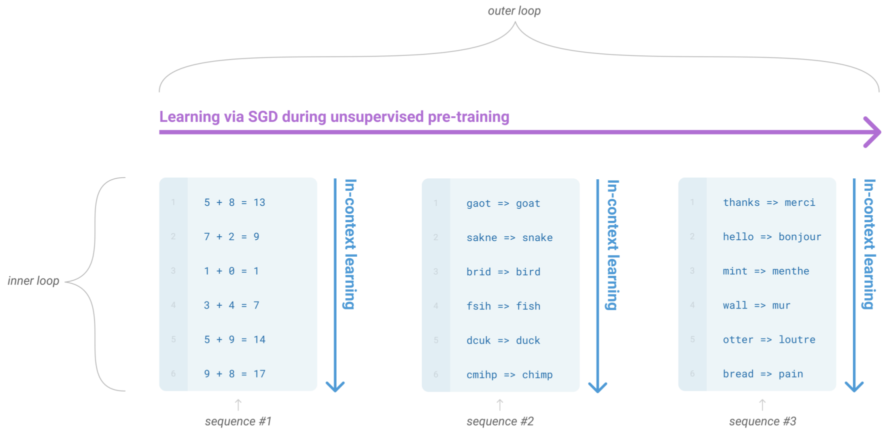
From Model Learning to Context Learning

Rethinking on Learning?

Learning by Model Update $\Longrightarrow$ In-Context Learning

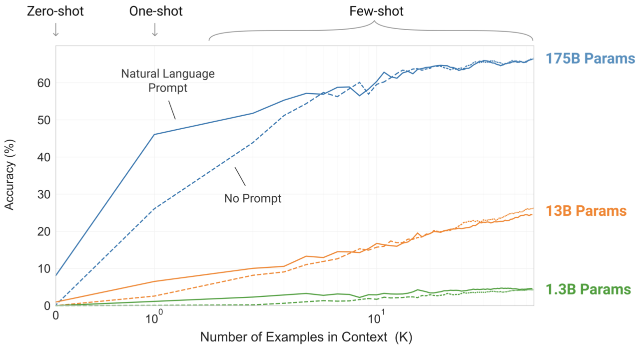
- Previous: update model parameters
- Part of current trends: preconditioning via prompting
  1. (observation) in-context learning empirically works [Brown et al., 2020]
  2. (interpretation) in-context learning $\approx$ gradient descent [Von Oswald et al., 2023]

Language Models are Few-Shot Learners [Brown et al., 2020]



- Outer loop: conventional model learning (*e.g.*, pretraining)
- Inner loop: in-context learning (*e.g.*, providing labeled examples in context)

Does In-Context Learning Work?



- Evaluated in toy tasks (*i.e.*, “character manipulation” tasks)
- Some interesting questions:
  - Why does it work?
  - When in-context learning work? (or any assumptions?)
  - Is the LLM adaptive to new tasks?

Why Does In-Context Learning Work? [Von Oswald et al., 2023]

Transformers Learn In-Context by Gradient Descent

Johannes von Oswald [1,2] **Eyvind Niklasson** [2] **Ettore Randazzo** [2] **João Sacramento** [1]
Alexander Mordvintsev [2] **Andrey Zhmoginov** [2] **Max Vladymyrov** [2]

- Interpret in-context learning theoretically
- In-context learning in Transformer forward passes ≈ gradient-based optimization (under some conditions)
  - "When training Transformers on auto-regressive tasks, in-context learning in the Transformer forward pass is implemented by gradient-based optimization of an implicit auto-regressive inner loss constructed from its in-context data."

Recall Gradient Descent

Regression Objective

$$L(W) = \frac{1}{2N} \sum_{i=1}^{N} \|Wx_i - y_i\|^2$$

- We assume the linear regression problem.
- W: parameters of a linear model
- N: the number of data points
- x_i: an example
- y_i: a label

One Step of Gradient Descent

Update Rule

$$W' \leftarrow W - \frac{\eta}{N} \sum_{i=1}^{N} r_i x_i^T \qquad \text{where} \qquad N \cdot \nabla_W L(W) = \sum_{i=1}^{N} \underbrace{(Wx_i - y_i)}_{r_i} x_i^T = \sum_{i=1}^{N} r_i x_i^T$$

- η: a learning rate
- r_i: residual – measures the degree of prediction error

Effect of One Step of GD on Prediction

Prediction

$$\hat{y}_j = W'x_j = \left(W - \eta' \sum_{i=1}^{N} r_i x_i^T\right) x_j = \underbrace{Wx_j}_{:=\hat{y}_j^{\text{old}}} \underbrace{-\eta' \sum_{i=1}^{N} r_i (x_i^T x_j)}_{\Delta \hat{y}_j}$$

- x_j: a new example
- $W' \leftarrow W - \eta' \sum_{i=1}^{N} r_i x_i^T$: one step of GD, where $\eta' = \frac{\eta}{N}$
- $\Delta \hat{y}_j$: prediction correction – this has an interesting structure

$$\sum_{i=1}^{N} r_i (x_i^T x_j) = \sum_{i=1}^{N} (\text{residual}) \times (\text{similarity})$$

  - (residual) $= Wx_i - y_i$
  - Ignoring η', the "new" prediction is corrected by prediction errors (=residual) on seen and similar examples.

Recall Attention

Linear Attention

$$\mathrm{Attn}(q) = \sum_i v_i(q^T k_i) = \sum_i (\text{value}) \times (\text{similarity})$$

- q: query – *e.g.*, an embedding vector of the current token
- k: key – *e.g.*, an id for a previous token's embedding vector
- v: value – *e.g.*, an embedding vector for the chosen key
- Here, assume that linear attention without Softmax – as you know, the following is the original attention:

$$\mathrm{Attn}(q) = \sum_i a_{i,j} v_j \qquad \text{where} \qquad a_{i,j} := \frac{\exp(q_i^T k_j)}{\sum_{j'} \exp(q_i^T k_{j'})}$$

Mapping Between Prediction Correction and Linear Attention

Mapping Trick

$$\underbrace{q = x_j}_{(=\text{current token})} \qquad \underbrace{k_i = x_i}_{(=\text{a previous token})} \qquad \underbrace{v_i = r_i = Wx_i - y_i}_{(=\text{residual of the previous token})}$$

- consider the following few shot prompt:

$$1 + 2 = 3$$
$$1 + 3 = 4$$
$$1 + 4 = ?$$

- $q = x_j$: *e.g.*, an embedding vector of "?" along with its context from previous tokens
- $k_i = x_i$: *e.g.*, a key embedding vector of "4" along with its context (*e.g.*, "$1 + 3 =$")
- $v_i = r_i$: *e.g.*, (an embedding for the prediction of 3) – (an embedding for 4)
  - assume that it is given, but can Transformer infer this?

Prediction Correction by GD = Linear Attention

Comparison

$$\Delta \hat{y}_j = \sum_i r_i (x_i^T x_j)$$
$$\mathrm{Attn}(x_j) = \sum_i r_i (x_j^T x_i)$$

- Exactly $\Delta \hat{y}_j = \mathrm{Attn}(x_j)$
- Any assumption?

Prediction Update by GD = Attention Residual Update

Comparison

$$
\begin{aligned}
&\text{(Prediction Update)} \quad \hat{y}_j = \hat{y}_j^{\text{old}} - \eta' \underbrace{\Delta \hat{y}_j}_{\text{correction}} \\
&\text{(Attention Update)} \quad h'_j = h_j + \underbrace{\text{Attn}(h_j)}_{\text{correction}}
\end{aligned}
$$

- What is the implication?
  - parameter update = attention rewiring?

Recap

- A very LLM is a meta learner.
  - In-context learning is working, empirically validated.
  - Actually, in-context learning in auto-regressive Transformer is very related to learning by one-step GD.
- Then, what is needed for learning agentic AI?
  - in-context learning?
  - context engineering?
  - post-training?
  - more few shots for in-context learning v.s. larger model?

Outline

Reference I

T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

J. Von Oswald, E. Niklasson, E. Randazzo, J. Sacramento, A. Mordvintsev, A. Zhmoginov, and M. Vladymyrov. Transformers learn in-context by gradient descent. In *International Conference on Machine Learning*, pages 35151–35174. PMLR, 2023.